

VRAAG / QUESTION 4

- (a) Wat is die verskil tussen die volgende tipes golwe?
 What is the difference between the following types of waves?
 Longitudinale golwe / Longitudinal waves.

Particles of medium are displaced in the same direction as the direction of the wave itself. (1)

Transversale golwe / Transverse waves.

Particles of the medium are displaced perpendicular to the direction of the wave itself. (1)

- (b) Wat is die verband tussen die periode T van 'n golf en sy frekwensie f ?
 Give the relation between the period T and the frequency f of a wave.

$$T = \frac{1}{f} \quad \text{OR} \quad f = \frac{1}{T}$$

- (c) Gee 'n uitdrukking vir die spoed v van 'n golf in terme van sy golflengte λ .
 Give the relation between the speed v of a wave and its wavelength λ .

$$v = \lambda f$$

- (d) 'n Bron met frekwensie f_0 beweeg met 'n spoed v_s na 'n stilstaande waarnemer. Indien die spoed van klank v is, bepaal:

A source with frequency f_0 is moving with speed v_s towards a receiver at rest. If the speed of sound is v , determine:

- (i) Die nuwe golflengte λ' . / The new wavelength λ' .

$$\lambda' = \lambda_0 - \frac{v_s}{f_0}$$

$$= \frac{v}{f_0} - \frac{v_s}{f_0}$$

$$\therefore \lambda' = \frac{v - v_s}{f_0}$$

- (ii) Bepaal die frekwensie wat die waarnemer hoor.

Determine the frequency observed by the receiver.

$$f' = \frac{v}{\lambda'} = \left(\frac{v}{v - v_s} \right) f_0$$

$$= f_0 \left(\frac{1}{1 - \frac{v_s}{v}} \right)$$